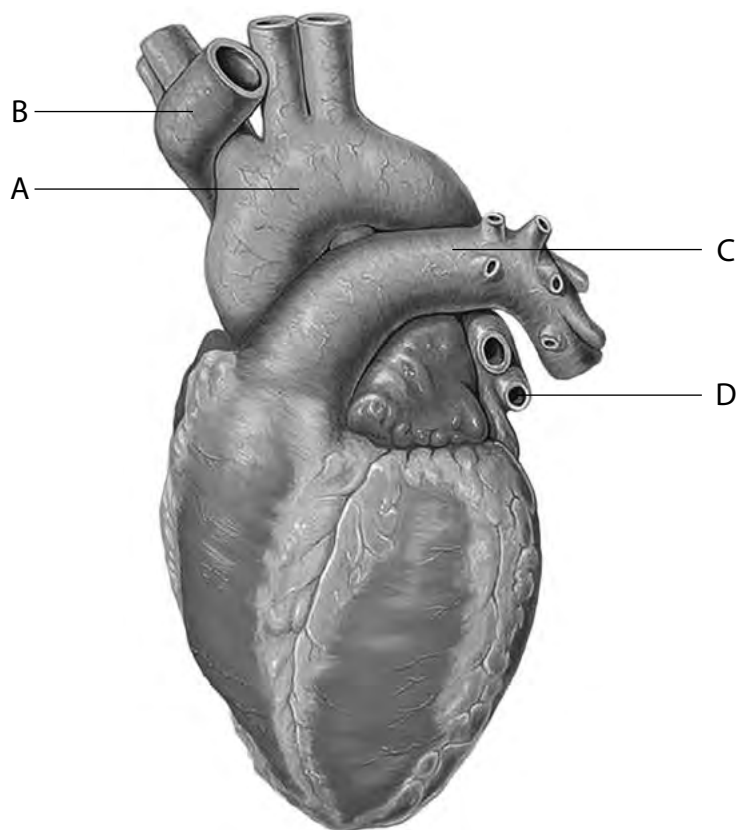


3 Mammals have a heart that pumps blood through a network of blood vessels.

(a) The drawing shows a human heart.



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(i) Which blood vessel takes blood from the heart to the body?

(1)

- ☐ A
- ☐ B
- ☐ C
- ☐ D

(ii) Which blood vessel has semilunar valves and contains blood with the highest concentration of carbon dioxide?

(1)

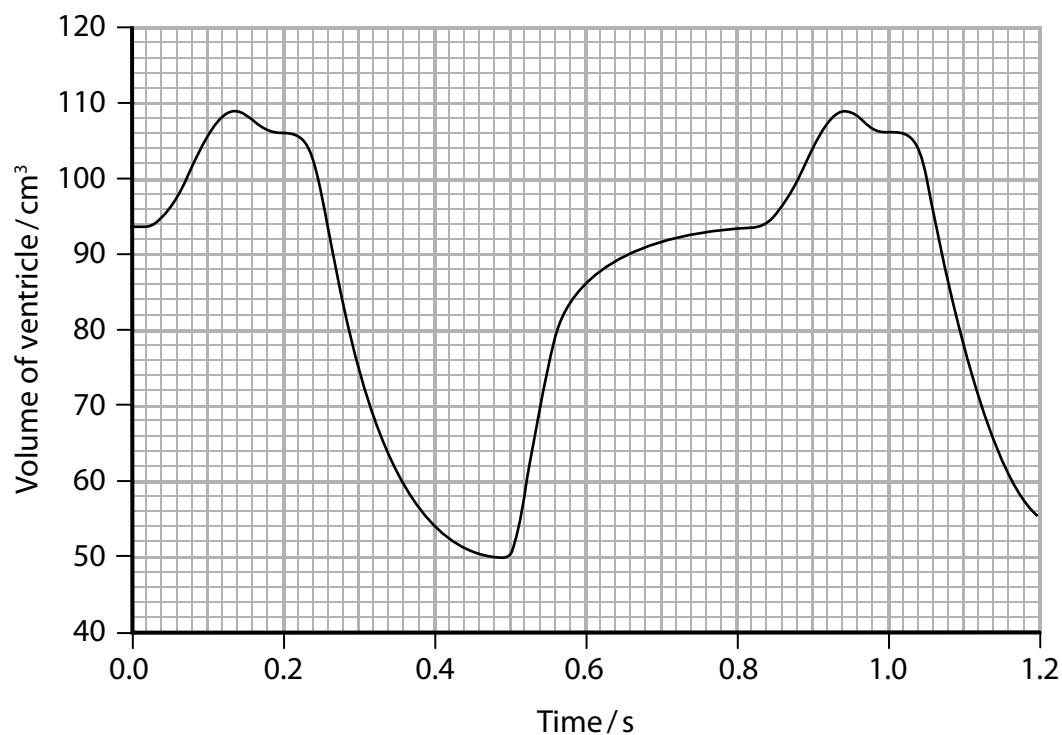
- ☐ A
- ☐ B
- ☐ C
- ☐ D

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- (b) The graph shows the change in volume of the left ventricle during the cardiac cycle.



- (i) When is this heart in ventricular systole?

(1)

- ☐ **A** at 0.1 seconds
- ☐ **B** at 0.4 seconds
- ☐ **C** at 0.6 seconds
- ☐ **D** at 0.8 seconds

- (ii) Calculate the volume of blood in dm^3 that will be pumped out of this heart by the left ventricle each minute.

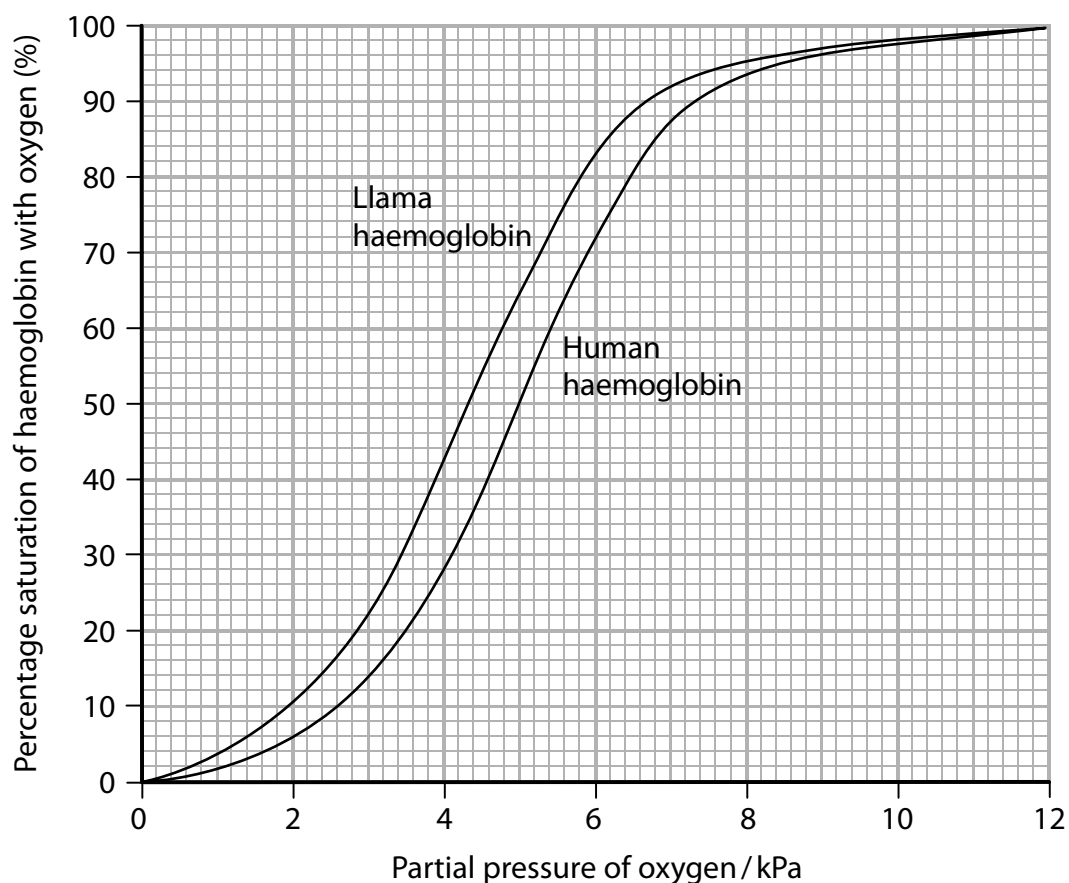
(3)

Answer dm^3

- (c) As altitude (height above sea level) increases, the partial pressure of oxygen in the air decreases.

Llamas are mammals that are adapted to living at high altitudes.

The graph shows the oxygen dissociation curves for llama haemoglobin and human haemoglobin.



Explain the differences between the dissociation curves. Use the information in the graph to support your answer.

(4)

(Total for Question 3 = 10 marks)